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The creative brain in the figural domain: Distinct patterns of EEG alpha power during idea generation and idea elaboration

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ABSTRACT

This study investigated EEG activity in the upper alpha band during the well-known Picture Completion Task of the Torrance Test of Creative Thinking (TTCT), a widely used creative ideation task in the figural domain. The application of a sophisticated computerized version of the TTCT facilitating the online assessment and digitalizing of participant's drawings allowed to separate two central stages of the creative ideation process (i.e., idea generation and idea elaboration). During idea generation, the participants' task was to generate an initial draft of an original and creative completion of the presented abstract lines and figures of the TTCT. During idea elaboration, the participants were required to mentally improve the originality of the initially generated idea/draft. Creative ideation in this figural task was generally associated with comparatively strong desynchronization of upper alpha power over parietal and occipital sites, indicating high visual/figural processing demands. Interestingly, the stage of idea elaboration was accompanied by a relative increase of upper alpha power at parietal and occipital sites compared to the stage of idea generation, indicating heightened top-down processing demands. Furthermore, task performance was associated with relative increases of upper alpha power at frontal sites and relative decreases at centro-temporal sites from the stage of idea generation to idea elaboration. This association suggests the importance of increased inhibitory control over stimulus-based bottom-up information and motor imagery in order to achieve more creative outputs. Taken together these findings add to the relevant literature in that they a) extend research on the relationship between EEG alpha activity and creativity to the figural domain, and b) support a multistage view of creative ideation, involving cognitive control and mental imagery as important components of creativity.

1. Introduction

Creativity is without any doubt one of the most important and fascinating skills of mankind, leading to numerous scientific studies trying to discover the underlying brain activation patterns associated with creative ideation (see, e.g., Arden et al., 2010; Jung and Vartanian, 2018). In recent years, the comparatively young field of neuroscientific research on creativity has made considerable process in unveiling relevant brain mechanisms associated with creative ideation by means of event-related (de)synchronization (ERD/ERS; Pfurtscheller, 1999) and task-related power changes (TRP; Gerloff, 1998) in the EEG alpha band (e.g., Benedek et al., 2011, 2014; Fink et al., 2007, 2009b; for an overview see Fink and Benedek, 2014).

A frequently reported finding is that creative ideation is associated with an increase of alpha power (relative to a resting baseline) at bilateral frontal and (right) posterior cortical sites, somewhat more pronounced in the upper alpha band (~ 10–12 Hz), which has been claimed as being more sensitive to specific task requirements (e.g., semantic memory processes, Klimesch, 1999; for an overview see Fink and Benedek, 2014). Interestingly, similar brain activation patterns were also reported for tasks that do not capture creative ideation in the narrower sense, like a word association task (Jauk et al., 2012), an open problem based reasoning task (Jaarsveld et al., 2015), during imagining dancing (Fink et al., 2009b), insight-full problem solving (Jung-Beeman et al., 2004), or creative cognitive reappraisal of a negative emotional event (Fink et al., 2017). The reported overlaps of brain activation patterns between creative ideation in the verbal domain and other tasks suggest that the observed pattern of TRP changes are, at least partly, independent of the task at hand and associated with cognitive functions (e.g., inhibitory, top-down control) implicated in many different creativity-related demands.

Task-related increases in the upper alpha band might strongly

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represent active inhibition of external stimuli, internal attention, and the shielding of working memory processes from task-irrelevant information (see, e.g., Fink and Benedek, 2014). The alpha power increase might reflect a mechanism increasing the signal to noise ratio, by attentional gating of task-relevant and task-irrelevant brain areas (Cooper et al., 2003; Crespo-Garcia et al., 2013; Foxe and Snyder, 2011; Jensen et al., 2002; Klimesch et al., 1999; Krause et al., 1996; Payne et al., 2013; Sauseng et al., 2005; von Stein and Sarnthein, 2000, for a short overview see Benedek, 2018).

Most of the electrophysiological knowledge about creative ideation/divergent thinking is based on verbal creative ideation tasks (e.g., the Alternate Uses Task, AUT; for an overview see, e.g., Arden et al., 2010; Fink and Benedek, 2014). However, if the reported electrophysiological activation patterns represent general mental processes relevant for creative ideation (e.g., top-down control, internal attention), studying creative ideation in other domains (e.g., figural, affective) should be associated with similar brain activation patterns than those observed during creative ideation in the verbal domain (for a similar approach see, e.g., Fink et al., 2017; Rominger et al., 2017b).

Former studies focusing on EEG correlates underlying figural creativity indicated a decrease of alpha power during creative ideation tasks compared to a reference period and to verbal creative ideation tasks (Jausovec and Jausovec, 2000; Petsche et al., 1997; Razumnikova et al., 2009; for review see Pidgeon et al., 2016). Though these studies considerably differ with respect to the applied methodological/analytical approaches and experimental settings from the studies focusing on verbal creative ideation, the reported pattern of task-related alpha power decreases during figural tasks seems to run counter the commonly found alpha power increases during verbal creative ideation (Fink and Benedek, 2014). The first aim of this study was thus to provide additional information regarding task-related alpha power changes during figural creative ideation and to explore whether creative ideation in the figural domain is associated with a similar pattern of findings reported in studies concerned with verbal creative ideation in applying a comparable experimental setting and analytical/methodological approach (see e.g., Fink et al., 2009a, Fink and Benedek, 2014). Furthermore, many authors proposed that creativity is a process with multiple stages (e.g., generative and exploratory, Finke et al., 1996; Benedek and Jauk, in press; Kleinmintz et al., 2017; Rominger et al., 2017a; Sowden et al., 2015). The study of Schwab et al. (2014) revealed that the process of thinking about original and alternative uses of objects went along with a characteristic time-course of TRP in the upper alpha band, reflecting distinct stages of the creative thinking process (see also Jaarsveld et al., 2015). The second aim of the present study, therefore, was to examine neurocognitive processes involved in different stages of creative ideation in the figural domain (i.e. idea generation and idea elaboration; for a similar approach see Ellamil et al., 2012).

In order to address these study questions, the well-known Picture Completion Task of the Torrance Test of Creative Thinking (TTCT; Torrance, 1966) was used. The TTCT provides optimal prerequisites to study brain activation during creative ideation in the figural domain. Firstly, by means of the TTCT it is possible to investigate different stages of the creative process, such as the stage of generating an original and creative idea and the stage of elaborating and improving the originality of this particular idea. These two stages are already implemented in the scoring procedure of the TTCT, which discriminates between both central features of divergent thinking, originality and elaboration of an idea (Gonen-Yaacovi et al., 2013; Torrance, 1966). Secondly, the TTCT shows high ecological validity, since drawing an original picture (or to complete a picture fragment creatively) is very close to every day creative performances in the figural domain (little c; Kaufman and Beghetto, 2009; see Runco et al., 2010). Specifically, in this study participants were instructed to complete the picture fragments as original and creative as possible during two separated stages. The participants' task was to generate an initial draft of an original and creative completion of the presented abstract lines and figures of the TTCT (stage of idea generation), which they had to further improve during the second stage of idea elaboration. In order to assess the electrophysiological activation during task performance adequately, both stages were divided into a phase of creative thinking/idea generation and a phase of active drawing.

We expected that the processes of idea generation and idea elaboration were associated with distinct patterns of alpha activity, which in turn should be associated with the performance outcome of the creative process (Fink et al., 2009a; Jausovec, 2000; Petsche et al., 1997; Rominger et al., 2017b).

2. Methods

2.1. Participants

Fifty individuals (48 university students, 32 women) participated in the study. The participants were recruited online via email and social media and offline via advertisements at the local institute. The age of the participants ranged from 19 to 42 ($M = 22.84$; $SD = 4.64$). All participants were right-handed (assessed by the hand dominance test, HDT; Steingrüber and Lienert, 1971; Papousek and Schulter, 1999), non-medicated, and written informed consent was obtained. Participants were requested to refrain from alcohol for twelve hours and from coffee and other stimulating beverages for two hours prior to their lab appointment, and to come to the session well rested. The study was approved by the local ethics committee.

2.2. TTCT

The Picture Completion Task of the TTCT (Torrance, 1966) is a well-known figural creativity/divergent thinking test (Kim, 2006). The participants' task is to complete or extend abstract lines or figures in a creative and original way. For the current study, the procedure of the TTCT was adjusted in order to be applicable during EEG recording and to allow to separate the stages of idea generation and idea elaboration. To this end, participants used a professional pen tablet (Wacom Intuos Pro Large), placed on a small table in front of them, to complete the picture fragments by means of a stylus connected to the tablet (see Fig. 1). The whole procedure of the TTCT was computerized by means of the software PsychoPy (Peirce, 2007; see Fig. 2). Computerizing the TTCT allowed the recording of the exact time-course and the digitalization of ideas and, hence, the differentiation between creative thinking/elaboration, active drawing, and the on-line presentation of the individually produced creative ideas.

Each trial started with a white cross (15 s), which served as reference period followed by the two different stages, the generation of the idea and the elaboration of this particular idea.

2.3. First stage – idea generation: Mental generation of an original and creative idea

During the first stage of figural idea generation, the participants were requested to mentally complete the picture fragments of the TTCT in an original and creative way. Therefore, each picture fragment was presented on the computer screen for 2 s (after the reference period), followed by an interrogation mark (idea generation, 10 s). Participants were instructed to carefully look at the picture fragment, and to start thinking creatively on it when the interrogation mark appeared. After the interrogation mark (stage of idea generation) participants were required to draw the mentally created idea and to complete the picture fragment, by means of a stylus connected to the professional pen tablet, during a fixed time window (20 s). Four seconds before the drawing period expired, a signal tone was presented. The digitalized drawing was saved automatically after completion and served as the visual input for the following second stage of idea elaboration.

Fig. 1. Application of professional pen tablet during the EEG-recording. (This figure is included with permission of pictured person.)

2.4. Second stage – idea elaboration: Mental elaboration and improvement of the originality of the previously generated idea/drawing

The stage of idea elaboration started with a white cross (3 s), followed by the presentation of the participants' previously generated idea (2 s). Then an interrogation mark was presented (10 s), indicating the participants' task to mentally elaborate and improve the originality of the initially generated idea/drawing. Following the interrogation mark, the participants' task was to add the elaborated ideas to the initially generated idea/drawing by means of the stylus connected to the professional pen tablet. Four seconds before this drawing phase ended, an auditory signal was delivered. The completed drawing was saved and the next trial started with a reference cross (schematic time-course see Fig. 2; similar procedure see, e.g., Fink et al., 2009a).

In order to gain a reliable measure of brain activation during the creation of an original picture, 10 additional picture fragments, resembling the original TTCT items, were constructed resulting in 30 items in total (Version A and B of the TTCT and 10 additionally constructed items, which were tested in a pre-study). All items were randomly presented in two blocks (15 items with a self-paced break in-between; minimum break of 3 min; about 45 min in total). In order to get used to the procedure, every participant explored the functions of the Wacom professional pen tablet during the time the EEG electrodes were attached. Additionally, each session started with an extensive explanation of the experiment's time course followed by three sample items.

2.5. Measures of creative performance in the TTCT

The produced drawings of the first stage (initially generated ideas) and the second stage (elaborated ideas) were rated by four independent raters on a four-point Likert scale ranging from "not original" to "very original". This procedure is a common approach in creativity research (cf. Consensual Assessment Technique; Amabile, 1982; in the figural domain see, e.g., Rominger et al., 2017a). The originality ratings showed good interrater reliability for the initially generated ideas (ICC = .76) and for the elaborated ideas (ICC = .77).

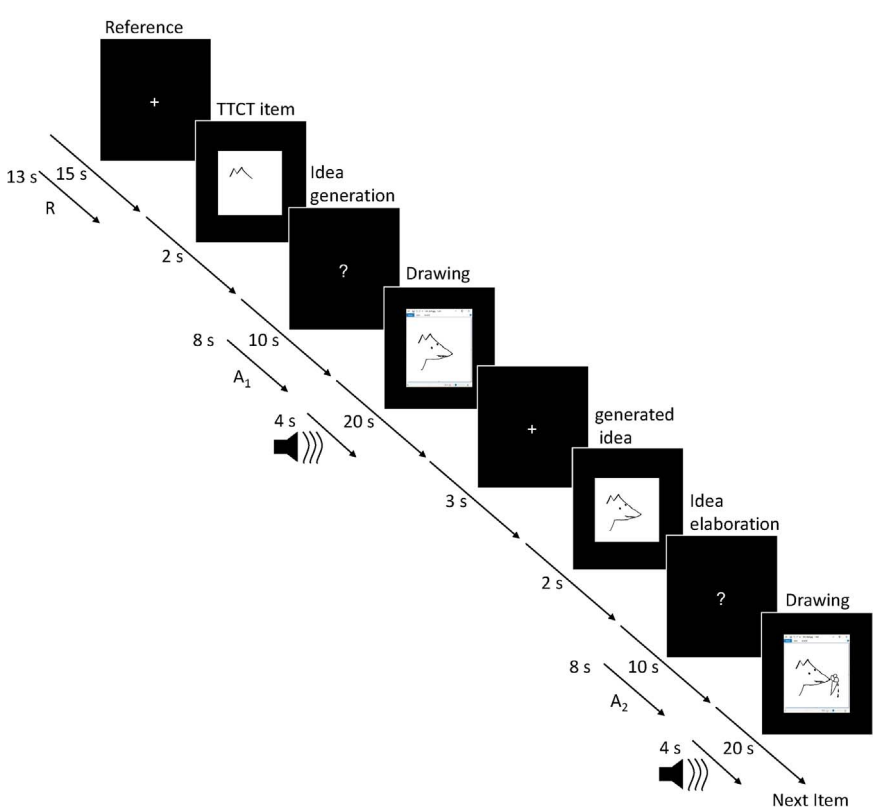

Fig. 2. Schematic time course of the computerized TTCT. A 13 s time period out of the 15 s fixation cross period served as the reference interval (R). A time period of 8 s out of 10 s served as the activation interval for the two phases of creative ideation, respectively (A_1, idea generation; A_2, idea elaboration). All TTCT items were shown for 2 s each.

2.6. EEG recordings and analysis

The EEG was recorded by 19 active electrodes, positioned according to the 10–20 system with a common EEG amplifier (Brainvision actiCHamp Research Amplifier, Brain Products™; 1000 Hz sampling rate, low cutoff DC, high cutoff 280 Hz) in a separate and quiet room. The ground electrode was located on the forehead, the reference electrodes on the nose. Vertical and horizontal electrooculograms (EOGs) were measured with two bipolar channels for horizontal and vertical eye movements. Electrode impedances were kept below 30 kΩ for all electrodes. The EEG signals were re-referenced to an averaged ears reference (Hagemann, 2004), down sampled to 500 Hz, and preprocessed by removing drifts and low pass filtering (50 Hz). The g.BSanalyze software (g.tec™, Graz, Austria) was used to manually check the resulting signal for artifacts (e.g., eye blinks or movements and muscle tension) and to calculate the band power values (μV^2) by squaring the filtered EEG signals. Only band power values of artifact free time-intervals were averaged (horizontally) by means of the median for each single trial and period. As in other studies concerning divergent thinking (e.g., Benedek et al., 2011; Fink et al., 2006), TRP scores were quantified for upper alpha power (10–12 Hz) for an electrode *i* by subtracting the log-transformed power of the reference period ($Pow_{i,\ reference}$) from the activation period ($Pow_{i,\ activation}$) for each trial *j* separately, according to the formula:

$$TRP_i = \text{Median}\left(\log(Pow_{i,\ activation})_j - \log(Pow_{i,\ reference})_j\right)$$

Negative values indicate a decrease of task-related alpha power from the reference to the activation period, while positive values express a power increase (Pfurtscheller and Lopes da Silva, 1999).

In addition to the classic analysis approach of task-related changes in alpha power (relative to a baseline), we calculated alpha power changes between the first (i.e., generation) and second stage (i.e., elaboration) of the creative thinking process. This trial-related approach might be especially sensitive to capture short-term process-related power changes (PRP, the change of activation from generation to elaboration within a trial, separately) in the upper alpha band, which in turn might be also associated with performance outcome. Comparing the TRP scores (median of power changes from baseline to activation) between the two stages (i.e., generation, elaboration) might possibly not adequately capture such fine-grained process related differences. Specifically, PRP change scores were calculated by subtracting the log-transformed power of the activation period of the first stage of idea generation ($Pow_{i,\ first\ idea}$) from the activation period of the second stage of idea elaboration ($Pow_{i,\ elaboration}$) for each trial *j* separately. Negative values of the PRP score indicate a relative decrease of upper alpha power from idea generation to idea elaboration.

For both analysis approaches, an 8 s interval from the 10 s activation periods for the first stage (idea generation) and the second stage (idea elaboration) served as activation intervals (1 s after the onset of the interrogation mark until 1 s before the offset). For the TRP analysis, a 13 s interval from the 15 s reference period served as reference interval (from 1 s after onset of the fixation cross until 1 s before the offset; see Fig. 2). The power changes between the reference and the activation period (TRP), as well as the power changes between both activation periods (PRP), in the upper alpha band (10–12 Hz), were computed for each electrode position and participant. Only activation periods and reference periods with at least 2 s artifact free EEG-recording were used for the statistical analyses. This procedure resulted in $M = 191.11$ s ($SD = 29.57$ s) artifact free EEG information during idea generation and $M = 192.73$ s ($SD = 25.81$ s) available EEG information during idea elaboration (reference: $M = 317.91$ s; $SD = 39.93$ s).

2.7. Statistical analysis

In order to evaluate if the two stages of the creative process during solving the TTCT (i.e., idea generation and the idea elaboration) differ with respect to task performance (originality), a paired samples *t*-test with the originality ratings of the two different stages of the creative process was performed.

Statistical analysis of EEG data was realized by an 8 × 2 × 2 ANOVA with the within-subjects factors AREA (eight electrode positions in each hemisphere), HEMISPHERE (left, right), and STAGE (idea generation, idea elaboration), and the TRP change scores as the dependent variable. The impact of task performance on the EEG results was investigated by additionally considering the continuous between-subjects variable originality of the elaborated ideas in the ANOVA design.

In order to explore more fine-grained associations between process-related power changes from the stage of idea generation to idea elaboration and task performance, an ANOVA with AREA and HEMISPHERE as within-subjects variables, the originality of the elaborated ideas as continuous between-subjects variable, and the PRP change scores as the dependent variable was run. For illustration of interaction effects between AREA and the continuous between-subjects variable, predicted PRP scores were calculated for one standard deviation below and one standard deviation above the sample mean of task performance (originality of elaborated ideas) using standard regression analysis.

In case of violations of sphericity assumptions, the multivariate approach was used (see Vasey and Thayer, 1987). Post-hoc comparisons were performed using Tukey's Honestly Significant Differences (HSD) tests. A significance level of $p < .05$ (two-tailed) was used for all analyses.

3. Results

3.1. Behavioral results

The paired samples *t*-test indicated that the elaborated pictures/drawings were rated as being more original ($M = 2.10$, $SD = .31$) than the pictures participants generated during the first stage of the experiment ($M = 1.54$, $SD = .25$; $t(49) = -25.88$, $p < .001$).

3.2. EEG results

3.2.1. Task-related changes in alpha power during figural creative ideation

The 8 × 2 × 2 ANOVA revealed a significant main effect AREA ($F(7,43) = 3.36$, $p = .006$, $\eta_p^2 = .11$), indicating decreased alpha power during figural creative ideation especially over posterior cortical sites (P7/8, see Fig. 3). The main effect HEMISPHERE ($F(1,49) = 10.47$, $p = .002$, $\eta_p^2 = .18$) indicated stronger alpha power decreases over left-

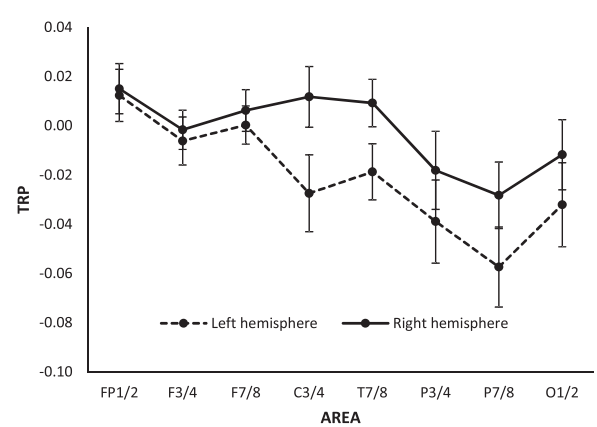

Fig. 3. General pattern of TRP and error bars (*SE*) in the upper alpha band during figural creative ideation.

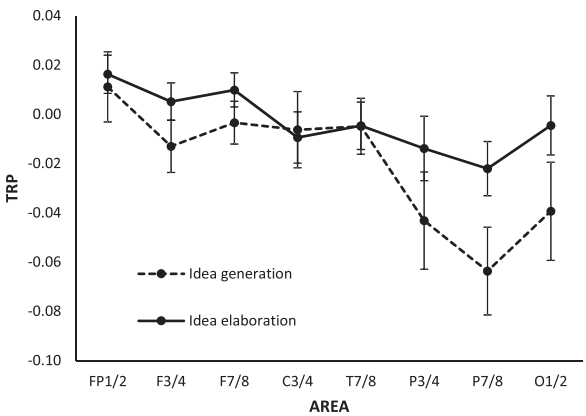

Fig. 4. TRP and error bars (*SE*) in the upper alpha band during the two different stages of figural creative ideation.

($M = -.021$, $SE = .009$) than right-hemispheric sites ($M = -.002$, $SE = .010$, see Fig. 3). The interaction AREA x HEMISPHERE was not significant ($F(7,43) = 1.19$, $p = .33$).

As evident in the significant main effect STAGE ($F(1,49) = 4.22$, $p = .045$, $\eta_p^2 = .08$), idea generation (M = − .020, SE = .012) was associated with stronger alpha power decreases than idea elaboration ($M = -.003$, $SE = .008$). The interaction AREA × STAGE ($F(7,43) = 3.57$, $p = .004$, $\eta_p^2 = .10$) revealed that the difference was driven by comparatively strong power decreases during idea generation (vs. elaboration) at parietal and occipital sites (post-hoc comparisons revealed significant STAGE related differences only at P3/4, P7/8, and O1/2; see Fig. 4). No further effect involving the within-subjects factor STAGE was significant.

3.2.2. *Alpha power changes during figural creative ideation as a function of task performance (originality of elaborated ideas)*

The ANOVA with task-related power changes in the upper alpha band as dependent variable revealed no significant effect involving the factor task performance. In contrast, the ANOVA with the process-related power changes in the upper alpha band (generation vs. elaboration) as dependent variable showed a significant interaction between the originality factor and AREA ($F(7,42) = 2.57$, $p = .027$, $\eta_p^2 = .04$). As illustrated in Fig. 5, originality in figural creative ideation was associated with relatively higher elaboration related increases of alpha power (relative to idea generation) over the frontal cortex (Fp1/2 and F3/4), and decreases over the central and temporal cortical sites (C3/4 and T7/8). No other effect involving the factor originality was significant.

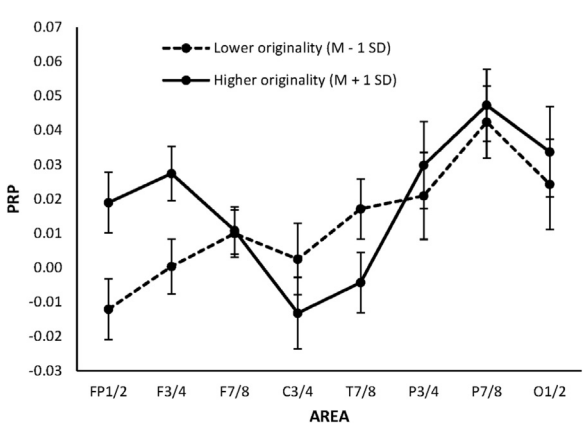

Fig. 5. PRP (process-related power chances from generation to elaboration) and error bars (*SE*) in the upper alpha band (positive values indicate an increase of power from the phase of idea generation to the phase of idea elaboration).

4. Discussion

Figural creative ideation was generally associated with comparatively strong decreases of alpha power at posterior cortical sites. These alpha power decreases were clearly more pronounced at the first stage of the creative thinking process while participants were required to generate a first draft of their ideas than at the second, elaboration stage. Elaborating on these initially drafted drawings resulted in a significantly lower decrease of alpha power (corresponds to relatively more alpha power during elaboration compared to generation) at posterior cortical sites. Individuals who generated more original drawings showed increased alpha power during elaboration relative to generation at frontal cortical sites and at the same time, decreased alpha power at centro-temporal cortical sites.

The overall pattern of findings clearly indicates that creative thinking in the figural domain is associated with relatively strong decreases in alpha power at parietal and occipital sites, reflecting high visual processing demands during the mental generation and manipulation of original and creative drawings. Generally, this finding is well in line with available EEG literature dealing with creative ideation in the figural domain (for review see Pidgeon et al., 2016). Petsche et al. (1997), for instance, reported a decrease of upper alpha during mental picture creation in comparison to a resting period (see also Razumnikova et al., 2009; Volf et al., 2010). In a similar vein, Jausovec and Jausovec (2000) found increased desynchronization in the upper alpha band during a picture completion task in contrast to a verbal creative ideation task. Consistent with these studies, the current finding of alpha power decreases at parieto-occipital sites supports the assumption that creative ideation in the figural domain (in this study predominantly during the stage of idea generation), is associated with various directly task-specific and sensory based visual mental operations, like visual imagery (Marks and Isaac, 1995; Petsche et al., 1997), mental rotation, spatial transformation, and imaginary manipulation of objects (Pfurtscheller et al., 1994; Riecanský and Katina, 2010; Sauseng et al., 2005; Williams and Rippon, 1995), as well as the imagination of movements (Erlacher et al., 2003; Pfurtscheller et al., 2006). In line with this, the originality of the drawings was associated with process related power decreases at centro-temporal sites from idea generation to idea elaboration (and at the same time power increases at frontal positions), which might further indicate the importance of motor imagery (i.e., imagine drawing a picture; Erlacher et al., 2003; Pfurtscheller et al., 2006) for creative ideation in the figural domain, especially during the stage of idea elaboration. This interpretation is also supported by recent fMRI studies, reporting the involvement of motor imagery (i.e., activation of premotor areas) during figural creativity (Aziz-Zadeh et al., 2013) as well as during music improvisation (Bengtsson et al., 2007), which, taken as a whole, nicely corresponds to the notion that imagination is highly important during creative ideation in general (Benedek, 2018; Finke et al., 1996; Jung et al., 2016).

However, verbal creative thinking (e.g., during the AUT) was found to be associated mainly with increased alpha power at frontal and posterior cortical sites, reflecting top-down control of sensory bottom-up stimulation and suppression of distracting information flow from the visual system (see, e.g., Crespo-Garcia et al., 2013; Fink and Benedek, 2014; Jensen et al., 2002; von Stein and Sarnthein, 2000). Figural creative ideation in contrast strongly relies on visual/figural mental operations, which are associated with decreased parieto-occipital alpha power (see, e.g., Pfurtscheller et al., 1994; Riecanský and Katina, 2010). Similar to Jausovec and Jausovec (2000), the divergence of neurophysiological activation patterns between creative ideation in the verbal and figural domain strongly suggests that the mental operations implicated in creative ideation are closely related to the specific contents (i.e., verbal vs. figural) and the specific targets (i.e., original association of alternative uses vs. original painting/drawing).

Nevertheless, nicely in line with current EEG literature in the verbal creativity domain, the findings of this study also clearly indicate the

importance of alpha power increases during creative ideation in the figural domain. The increase of originality from idea generation to idea elaboration was paralleled by a relative increase of alpha power over parieto-occipital sites. In addition, the higher the originality of the generated drawings, the higher was the relative increase of alpha from idea generation to idea elaboration at prefrontal sites. In line with Jaarsveld et al. (2015) and Schwab et al. (2014), who reported increased alpha power at later stages of the creative thinking process, these findings may indicate heightened internal processing demands and top-down control of external stimuli (Fink and Benedek, 2014) during the later stage of idea elaboration/evaluation (compared to idea generation; see e.g., Ellamil et al., 2012). This further supports the notion that different stages of the creative thinking process are associated with different patterns of brain activity (see also Ellamil et al., 2012). Particularly the involvement of prefrontal areas in creative ideation is a well-known finding in different domains (verbal: Abraham et al., 2012; Fink et al., 2009a, 2010; figural: Aziz-Zadeh et al., 2013; de Souza et al., 2010; Ellamil et al., 2012; Huang et al., 2013; Kowatari et al., 2009; Lustenberger et al., 2015; Shamay-Tsoory et al., 2011; musical: Lopata et al., 2017) and different contexts (e.g., affective: Fink et al., 2017; Papousek et al., 2017; Perchtold et al., 2018a, 2018b), which strongly indicates the involvement of executive and cognitive control processes in various creativity related task demands. Especially at later stages of the creative thinking process, when individuals are evaluating and elaborating initially generated ideas, more intelligent related styles of thinking might be relevant (Ellamil et al., 2012; Jaarsveld et al., 2015).

Nevertheless, the current approach has also some important limitations. Firstly, this study did not directly compare EEG activation patterns during creative ideation in the figural and the verbal domain. Future studies in this field are thus particularly challenged to examine the issue of domain specificity vs. generality of creative ideation within one and the same experimental design. Another possible drawback of the applied methodological approach is that only two stages of the creative thinking process were examined. While idea generation and elaboration may certainly represent two important components of the creative thinking process, this somewhat artificial experimental setting only reflects a limited picture of the manifold processes involved in creativity. Future research should hence also attempt to investigate creative behavior in more ecologically valid creativity tasks involving more real-life creativity demands.

Taken together, the observed pattern of associations argues for a multistage view of creative ideation, where executive control (e.g., Benedek et al., 2012; Fink et al., 2018) and mental imagery (e.g., elaboration and manipulation of objects) are necessary to achieve a creative outcome (e.g., generative and explorative/evaluative phase of the geneplore model; Finke et al., 1996; see also Ellamil et al., 2012). The neurophysiological findings indicate that the process of drawing/imaging a creative and original picture might need a high degree of elaboration, composing, planning, and the imagery of concrete implementation of generated ideas and objects into an eventual creative composition.

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Author note

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References

Abraham, A., Pieritz, K., Thybusch, K., Rutter, B., Kröger, S., Schweckendiek, J., Stark, R., Windmann, S., Hermann, C., 2012. Creativity and the brain: uncovering the neural signature of conceptual expansion. Neuropsychologia 50, 1906–1917. http://dx.doi.org/10.1016/j.neuropsychologia.2012.04.015.

Amabile, T.M., 1982. Social psychology of creativity: a consensual assessment technique. J. Personal. Social. Psychol. 43, 997–1013. http://dx.doi.org/10.1037/0022-3514.43.5.997.

Arden, R., Chavez, R.S., Grazioplene, R., Jung, R.E., 2010. Neuroimaging creativity: a psychometric view. Behav. Brain Res. 214, 143–156. http://dx.doi.org/10.1016/j.bbr.2010.05.015.

Aziz-Zadeh, L., Liew, S.-L., Dandekar, F., 2013. Exploring the neural correlates of visual creativity. Social. Cogn. Affect. Neurosci. 8, 475–480. http://dx.doi.org/10.1093/scan/nss021.

Benedek, M., 2018. Internally directed attention in creative cognition. In: Jung, R.E., Vartanian, O. (Eds.), The Cambridge Handbook of the Neuroscience of Creativity. Cambridge University Press, New York, pp. 180–194. http://dx.doi.org/10.1017/9781316556238.011.

Benedek M., and Jauk E., Spontaneous and controlled processes in creative cognition. In: Fox K.C.R., & Christoff K., (Eds.), The Oxford Handbook of Spontaneous Thought: Mind-wandering, Creativity, Dreaming, and Clinical Conditions, Oxford University Press; New York (in press).

Benedek, M., Bergner, S., Könen, T., Fink, A., Neubauer, A.C., 2011. EEG alpha synchronization is related to top-down processing in convergent and divergent thinking. Neuropsychologia 49, 3505–3511. http://dx.doi.org/10.1016/j.neuropsychologia.2011.09.004.

Benedek, M., Franz, F., Heene, M., Neubauer, A.C., 2012. Differential effects of cognitive inhibition and intelligence on creativity. Personal. Individ. Differ. 53, 480–485. http://dx.doi.org/10.1016/j.paid.2012.04.014.

Benedek, M., Schickel, R.J., Jauk, E.V., Fink, A., Neubauer, A.C., 2014. Alpha power increases in right parietal cortex reflects focused internal attention. Neuropsychologia 56, 393–400. http://dx.doi.org/10.1016/j.neuropsychologia.2014.02.010.

Bengtsson, S.L., Csíkszentmihályi, M., Ullén, F., 2007. Cortical regions involved in the generation of musical structures during improvisation in pianists. J. Cogn. Neurosci. 19, 830–842. http://dx.doi.org/10.1162/jocn.2007.19.5.830.

Cooper, N.R., Croft, R.J., Dominey, S.J.J., Burgess, A.P., Gruzelier, J.H., 2003. Paradox lost? Exploring the role of alpha oscillations during externally vs. internally directed attention and the implications for idling and inhibition hypotheses. Int. J. Psychophysiol. 47, 65–74. http://dx.doi.org/10.1016/S0167-8760(02)00107-1.

Crespo-Garcia, M., Pinal, D., Cantero, J.L., Díaz, F., Zurrón, M., Atienza, M., 2013. Working memory processes are mediated by local and long-range synchronization of alpha oscillations. J. Cogn. Neurosci. 25, 1343–1357. http://dx.doi.org/10.1162/jocn_a_00379.

Ellamil, M., Dobson, C., Beeman, M., Christoff, K., 2012. Evaluative and generative modes of thought during the creative process. NeuroImage 59, 1783–1794. http://dx.doi.org/10.1016/j.neuroimage.2011.08.008.

Erlacher, D., Schredl, M., LaBerge, S., 2003. Motor area activation during dreamed hand clenching: a pilot study on EEG alpha band. Sleep Hypn. 5, 182–187.

Fink, A., Benedek, M., 2014. EEG alpha power and creative ideation. Neurosci. Biobehav. Rev. 44, 111–123. http://dx.doi.org/10.1016/j.neubiorev.2012.12.002.

Fink, A., Grabner, R.H., Benedek, M., Neubauer, A.C., 2006. Divergent thinking training is related to frontal electroencephalogram alpha synchronization. Eur. J. Neurosci. 23, 2241–2246. http://dx.doi.org/10.1111/j.1460-9568.2006.04751.x.

Fink, A., Benedek, M., Grabner, R.H., Staudt, B., Neubauer, A.C., 2007. Creativity meets neuroscience: experimental tasks for the neuroscientific study of creative thinking. Methods 42, 68–76. http://dx.doi.org/10.1016/j.ymeth.2006.12.001.

Fink, A., Grabner, R.H., Benedek, M., Reishofer, G., Hauswirth, V., Fally, M., Neuper, C., Ebner, F., Neubauer, A.C., 2009a. The creative brain: investigation of brain activity during creative problem solving by means of EEG and fMRI. Hum. Brain Mapp. 30, 734–748. http://dx.doi.org/10.1002/hbm.20538.

Fink, A., Graif, B., Neubauer, A.C., 2009b. Brain correlates underlying creative thinking: EEG alpha activity in professional vs. novice dancers. NeuroImage 46, 854–862. http://dx.doi.org/10.1016/j.neuroimage.2009.02.036.

Fink, A., Grabner, R.H., Gebauer, D., Reishofer, G., Koschutnig, K., Ebner, F., 2010. Enhancing creativity by means of cognitive stimulation: evidence from an fMRI study. NeuroImage 52, 1687–1695. http://dx.doi.org/10.1016/j.neuroimage.2010.05.072.

Fink, A., Weiss, E.M., Schwarzl, U., Weber, H., Assuncao, V.L., de, Rominger, C., Schulter, G., Lackner, H.K., Papousek, I., 2017. Creative ways to well-being: reappraisal inventiveness in the context of anger-evoking situations. Cogn. Affect. Behav. Neurosci. 17, 94–105. http://dx.doi.org/10.3758/s13415-016-0465-9.

Fink, A., Perchtold, C.M., Rominger, C., 2018. Creativity and cognitive control in the Cognitive and Affective Domains. In: Jung, R.E., Vartanian, O. (Eds.), The Cambridge Handbook of the Neuroscience of Creativity. Cambridge University Press, New York, pp. 318–332. http://dx.doi.org/10.1017/9781316556238.019.

Finke, R.A., Ward, T.B., Smith, S.M., 1996. Creative Cognition: Theory, Research, and Applications. MIT Press, Cambridge, MA.

Foxe, J.J., Snyder, A.C., 2011. The role of alpha-band brain oscillations as a sensory suppression mechanism during selective attention. Front. Psychol. 2, 154. http://dx.doi.org/10.3389/fpsyg.2011.00154.

Gerloff, C., 1998. Functional coupling and regional activation of human cortical motor areas during simple, internally paced and externally paced finger movements. Brain 121, 1513–1531. http://dx.doi.org/10.1093/brain/121.8.1513.

Gonen-Yaacovi, G., de Souza, L.C., Levy, R., Urbanski, M., Josse, G., Volle, E., 2013. Rostral and caudal prefrontal contribution to creativity: a meta-analysis of functional imaging data. Front. Hum. Neurosci. 7, 465. http://dx.doi.org/10.3389/fnhum.2013.00465.

Hagemann, D., 2004. Individual differences in anterior EEG asymmetry: methodological problems and solutions. Biol. Psychol. 67, 157–182. http://dx.doi.org/10.1016/j.biopsycho.2004.03.006.
Huang, P., Qiu, L., Shen, L., Zhang, Y., Song, Z., Qi, Z., Gong, Q., Xie, P., 2013. Evidence for a left-over-right inhibitory mechanism during figural creative thinking in healthy nonartists. Hum. Brain Mapp. 34, 2724–2732. http://dx.doi.org/10.1002/hbm.22093.
Jaarsveld, S., Fink, A., Rinner, M., Schwab, D., Benedek, M., Lachmann, T., 2015. Intelligence in creative processes: an EEG study. Intelligence 49, 171–178. http://dx.doi.org/10.1016/j.intell.2015.01.012.
Jauk, E.V., Benedek, M., Neubauer, A.C., 2012. Tackling creativity at its roots: evidence for different patterns of EEG alpha activity related to convergent and divergent modes of task processing. Int. J. Psychophysiol. 84, 219–225. http://dx.doi.org/10.1016/j.ijpsycho.2012.02.012.
Jausovec, N., 2000. Differences in cognitive processes between gifted, intelligent, creative, and average individuals while solving complex problems: an EEG study. Intelligence 28, 213–237.
Jausovec, N., Jausovec, K., 2000. EEG activity during the performance of complex mental problems. Int. J. Psychophysiol. 36, 73–88.
Jensen, O., Gelfand, J., Kounios, J., Lisman, J.E., 2002. Oscillations in the alpha band (9–12 Hz) increase with memory load during retention in a short-term memory task. Cereb. Cortex 12, 877–882. http://dx.doi.org/10.1093/cercor/12.8.877.
Jung R.E. and Vartanian O. (Eds.), 2018. The Cambridge Handbook of the Neuroscience of Creativity, Cambridge University Press; New York.
Jung, R.E., Flores, R.A., Hunter, D., 2016. A new measure of imagination ability: anatomical brain imaging correlates. Front. Psychol. 7, 496. http://dx.doi.org/10.3389/fpsyg.2016.00496.
Jung-Beeman, M., Bowden, E.M., Haberman, J., Frymiare, J.L., Arambel-Liu, S., Greenblatt, R., Reber, P.J., Kounios, J., 2004. Neural activity when people solve verbal problems with insight. PLoS Biol. 2, e97.
Kaufman, J.C., Beghetto, R.A., 2009. Beyond big and little: the four C model of creativity. Rev. Gen. Psychol. 13, 1–12. http://dx.doi.org/10.1037/a0013688.
Kim, K.H., 2006. Can we trust creativity tests? A review of the Torrance Tests of Creative Thinking (TTCT). Creat. Res. J. 18, 3–14. http://dx.doi.org/10.1207/s15326934crj1801_2.
Kleinmintz, O.M., Abecasis, D., Tauber, A., Geva, A., Chistyakov, A.V., Kreinin, I., Klein, E., Shamay-Tsoory, S.G., 2017. Participation of the left inferior frontal gyrus in human originality. Brain Struct. Funct. http://dx.doi.org/10.1007/s00429-017-1500-5.
Klimesch, W., 1999. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. Brain Res. Rev. 29, 169–195. http://dx.doi.org/10.1016/S0165-0173(98)00056-3.
Klimesch, W., Doppelmayr, M., Schwaiger, J., Auinger, P., Winkler, T., 1999. 'Paradoxical' alpha synchronization in a memory task. Cogn. Brain Res. 7, 493–501. http://dx.doi.org/10.1016/S0926-6410(98)00056-1.
Kowatari, Y., Lee, S.H., Yamamura, H., Nagamori, Y., Levy, P., Yamane, S., Yamamoto, M., 2009. Neural networks involved in artistic creativity. Hum. Brain Mapp. 30, 1678–1690. http://dx.doi.org/10.1002/hbm.20633.
Krause, C.M., Lang, H.A., Laine, M., Kuusisto, M., Pörn, B., 1996. Event-related. EEG desynchronization and synchronization during an auditory memory task. Electroencephalogr. Clin. Neurophysiol. 98, 319–326. http://dx.doi.org/10.1016/0013-4694(96)00283-0.
Lopata, J.A., Nowicki, E.A., Joanisse, M.F., 2017. Creativity as a distinct trainable mental state: an EEG study of musical improvisation. Neuropsychologia 99, 246–258. http://dx.doi.org/10.1016/j.neuropsychologia.2017.03.020.
Lustenberger, C., Boyle, M.R., Foulser, A.A., Mellin, J.M., Fröhlich, F., 2015. Functional role of frontal alpha oscillations in creativity. Cortex 67, 74–82. http://dx.doi.org/10.1016/j.cortex.2015.03.012.
Marks, D.F., Isaac, A.R., 1995. Topographical distribution of EEG activity accompanying visual and motor imagery in vivid and non-vivid imagers. Br. J. Psychol. 86, 271–282. http://dx.doi.org/10.1111/j.2044-8295.1995.tb02561.x.
Papousek, I., Schulter, G., 1999. Quantitative assessment of five behavioural laterality measures: distributions of scores and intercorrelations among right-handers. Laterality 4, 345–362. http://dx.doi.org/10.1080/713754344.
Papousek, I., Weiss, E.M., Perchtold, C.M., Weber, H., Assuncao, V.L., de, Schulter, G., Lackner, H.K., Fink, A., 2017. The capacity for generating cognitive reappraisals is reflected in asymmetric activation of frontal brain regions. Brain Imaging Behav. 11, 577–590. http://dx.doi.org/10.1007/s11682-016-9537-2.
Payne, L., Guillory, S., Sekuler, R., 2013. Attention-modulated alpha-band oscillations protect against intrusion of irrelevant information. J. Cogn. Neurosci. 25, 1463–1476. http://dx.doi.org/10.1162/jocn_a_00395.
Peirce, J.W., 2007. PsychoPy-Psychophysics software in Python. J. Neurosci. Methods 162, 8–13. http://dx.doi.org/10.1016/j.jneumeth.2006.11.017.
Perchtold, C.M., Fink, A., Rominger, C., Weber, H., Loureiro de Assunção, V., Schulter, G., Weiss, E.M., Papousek, I., 2018a. Reappraisal inventiveness: Impact of appropriate brain activation during efforts to generate alternative appraisals on the perception of chronic stress in women. Anxiety Stress Coping 31, 206–221. http://dx.doi.org/10.1080/10615806.2017.1419205.
Perchtold, C.M., Papousek, I., Koschutnig, K., Rominger, C., Weber, H., Weiss, E.M., Fink, A., 2018b. Affective creativity meets classic creativity in the scanner. Hum. Brain Mapp. 39, 393–406. http://dx.doi.org/10.1002/hbm.23851.
Petsche, H., Kaplan, S., Stein, A. von, Filz, O., 1997. The possible meaning of the upper and lower alpha frequency ranges for cognitive and creative tasks. Int. J. Psychophysiol. 26, 77–97. http://dx.doi.org/10.1016/S0167-8760(97)00757-5.
Pfurtscheller, G., 1999. Quantification of ERD and ERS in the time domain. In: Pfurtscheller, G., Lopes da Silva, F.H. (Eds.), Event-related Desynchronization. Elsevier, Amsterdam, pp. 89–105.
Pfurtscheller, G., Lopes da Silva, F.H., 1999. Event-related EEG/MEG synchronization and desynchronization: basic principles. Clin. Neurophysiol. 110, 1842–1857.
Pfurtscheller, G., Neuper, C., Mohl, W., 1994. Event-related desynchronization (ERD) during visual processing. Int. J. Psychophysiol. 16, 147–153. http://dx.doi.org/10.1016/0167-8760(89)90041-X.
Pfurtscheller, G., Brunner, C., Schlögl, A., Lopes da Silva, F.H., 2006. Mu rhythm (de) synchronization and EEG single-trial classification of different motor imagery tasks. NeuroImage 31, 153–159. http://dx.doi.org/10.1016/j.neuroimage.2005.12.003.
Pidgeon, L.M., Grealy, M., Duffy, A.H.B., Hay, L., McTeague, C., Vuletic, T., Gilbert, S.J., 2016. Functional neuroimaging of visual creativity: a systematic review and meta-analysis. Brain Behav. e00540. http://dx.doi.org/10.1002/brb3.540.
Razumnikova, O.M., Volf, N.V., Tarasova, I.V., 2009. Strategy and results: sex differences in electrographic correlates of verbal and figural creativity. Hum. Physiol. 35, 285–294. http://dx.doi.org/10.1134/S0362119709030049.
Riecanský, I., Katina, S., 2010. Induced EEG alpha oscillations are related to mental rotation ability: the evidence for neural efficiency and serial processing. Neurosci. Lett. 482, 133–136. http://dx.doi.org/10.1016/j.neulet.2010.07.017.
Rominger, C., Fink, A., Weiss, E.M., Bosch, J., Papousek, I., 2017a. Allusive thinking (remote associations) and auditory top-down inhibition skills differentially predict creativity and positive schizotypy. Cogn. Neuropsychiatry 22, 108–121. http://dx.doi.org/10.1080/13546805.2016.1278361.
Rominger, C., Reitinger, J., Seyfried, C., Schneckenleitner, E., Fink, A., 2017b. The reflecting brain: reflection competence in an educational setting is associated with increased electroencephalogram activity in the alpha band. Mind Brain Educ. 49, 3505. http://dx.doi.org/10.1111/mbe.12140.
Runco, M.A., Millar, G., Acar, S., Cramond, B., 2010. Torrance Tests of Creative Thinking as predictors of personal and public achievement: a fifty-year follow-up. Creat. Res. J. 22, 361–368. http://dx.doi.org/10.1080/10400419.2010.523393.
Sauseng, P., Klimesch, W., Doppelmayr, M., Pecherstorfer, T., Freunberger, R., Hanslmayr, S., 2005. EEG alpha synchronization and functional coupling during top-down processing in a working memory task. Hum. Brain Mapp. 26, 148–155. http://dx.doi.org/10.1002/hbm.20150.
Schwab, D., Benedek, M., Papousek, I., Weiss, E.M., Fink, A., 2014. The time-course of EEG alpha power changes in creative ideation. Front. Hum. Neurosci. 8, 310. http://dx.doi.org/10.3389/fnhum.2014.00310.
Shamay-Tsoory, S.G., Adler, N., Aharon-Peretz, J., Perry, D., Mayseless, N., 2011. The origins of originality: the neural bases of creative thinking and originality. Neuropsychologia 49, 178–185.
de Souza, L.C., Volle, E., Bertoux, M., Czernecki, V., Funkiewiez, A., Allali, G., Leroy, B., Sarazin, M., Habert, M.-O., Dubois, B., Kas, A., Levy, R., 2010. Poor creativity in frontotemporal dementia: a window into the neural bases of the creative mind. Neuropsychologia 48, 3733–3742. http://dx.doi.org/10.1016/j.neuropsychologia.2010.09.010.
Sowden, P.T., Pringle, A., Gabora, L., 2015. The shifting sands of creative thinking: connections to dual-process theory. Think. Reason. 21, 40–60. http://dx.doi.org/10.1080/13546783.2014.885464.
von Stein, A., Sarnthein, J., 2000. Different frequencies for different scales of cortical integration: from local gamma to long range alpha/theta synchronization. Int. J. Psychophysiol. 38, 301–313. http://dx.doi.org/10.1016/S0167-8760(00)00172-0.
Steingrüber, H.-J., Lienert, G., 1971. Hand-Dominanz-Test. Hogrefe, Göttingen.
Torrance, P.E., 1966. Torrance Test of Creative Thinking: Norms-technical Manual Research Edition Verbal Tests, Forms A and B Figural Tests, Forms A and B. Personnel Press, Princeton, New Jersey.
Vasey, M.W., Thayer, J.F., 1987. The continuing problem of false positives in repeated measures ANOVA in psychophysiology: a multivariate solution. Psychophysiology 24, 479–486.
Volf, N.V., Tarasova, I.V., Razumnikova, O.M., 2010. Motivation in creative task: impact on individual differences in task-related alpha oscillations. In: Proceedings of the Advances in Biomedical Research: Proceedings of the International Conference on Medical Physiology, pp. 215–221.
Williams, J.D., Rippon, G., 1995. Psychophysiological correlates of dynamic imagery. Br. J. Psychol. 86, 283–300.